

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0202

Tellico 138 kV
March 25, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0202. The Interconnection Customer proposes a 286.3 MW wind facility interconnecting at the Tellico 138 kV substation in the Kentucky area. The request was received on February 02, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 151 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$239,826,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Greenwater – Ss3144 Tap 138 kV, the Tellico – Eldorado 138 kV loads to 191.6% of its 198 MVA rating. Project DFAX: 100.0%.

For the loss of Eldorado – Tellico 138 kV, the Greenwater – Ss3144 Tap 138 kV loads to 163.0% of its 233 MVA rating. Project DFAX: 100.0%.

The Twisp – Vly Springs 138 kV is overloaded to 132.0% of its 192 MVA emergency rating for the loss of the Eldorado – Tellico 138 kV. The project contributes 100.0% DFAX (286.3 MW) to this overload.

For the loss of Eldorado – Tellico 138 kV, the Alder Creek-Bradshaw Substation – Tracy Defense Depot Substation 138 kV loads to 131.9% of its 233 MVA rating. Project DFAX: 57.6%.

The La Fresa – Munk 138 kV is overloaded to 131.8% of its 215 MVA emergency rating for the loss of the Eldorado – Tellico 138 kV. The project contributes 57.6% DFAX (164.9 MW) to this overload.

For the loss of Eldorado – Tellico 138 kV L, the Alder Creek-Bradshaw – La Fresa 138 kV loads to 131.5% of its 225 MVA rating. Project DFAX: 57.6%.

The Wissota – Berry St 138 kV line is overloaded to 131.5% of its 272 MVA emergency rating for the loss of the Killen – Toledo Bend 500 kV. The project contributes 11.9% DFAX (34.0 MW) to this overload.

Norwich – Larson Street 230 kV L: 131.0% loading (668 MVA) following the outage of the Barrett – Norwich Substation 500 kV L; DFAX = 10.0%.

Firstenergy Albright – Munk Substation 138 kV line: 130.7% loading (199 MVA) following the outage of the Tellico – Eldorado 138 kV; DFAX = 57.6%.

For the loss of Barrett – Norwich 500 kV, the Minco – Larson Street 230 kV loads to 116.8% of its 786 MVA rating. Project DFAX: 14.8%.

Wnthrop – Minco 230 kV Line: 114.2% loading (546 MVA) following the outage of the Barrett – Norwich 500 kV L; DFAX = 5.8%.

Goldsboro – Ironman 230 kV: 110.6% loading (402 MVA) following the outage of the Toledo Bend – Killen 500 kV; DFAX = 7.9%.

For the loss of Piney Crk Project Substation – Hilltop 500 kV, the Piney Crk Project 500 kV/345 kV XFMR loads to 109.2% of its 1,257 MVA rating. Project DFAX: 8.6%.

The Gavin Power Llc – Olympus 230 kV is overloaded to 108.7% of its 611 MVA emergency rating for the loss of the Gavin Power Llc 500/345kV Autotransformer. The project contributes 8.1% DFAX (23.3 MW) to this overload.

For the loss of Little Chute – Ft Humbug 345 kV, the Fort Humbug – Coronet 500 kV L loads to 106.9% of its 1,936 MVA rating. Project DFAX: 9.4%.

Allegheny Energy Units 3 4 & 5 345/1kV TX: 106.9% loading (951 MVA) following the outage of the Hilltop – Piney Creek Project 500 kV; DFAX = 9.6%.

The Allegheny Energy Units 3 4 & 5 500/1 kV Transformer is overloaded to 106.7% of its 953 MVA emergency rating for the loss of the Hilltop Substation – Piney Crk Project 500 kV. The project contributes 9.6% DFAX (27.5 MW) to this overload.

For the loss of Hilltop – Talenenergy Martins Creek LlcWilliamsport 500 kV, the Hilltop 500 kV/345 kV XFMR loads to 106.6% of its 1,385 MVA rating. Project DFAX: 7.5%.

Blackrock – Shawville Station 345 kV: 105.9% loading (1,099 MVA) following the outage of the Carbon Center 500/345 kV Transformer Bank; DFAX = 7.8%.

Shawville Sta 500/345 kV Transformer: 104.2% loading (724 MVA) following the outage of the Carbon Center 500-345 kV Transformer; DFAX = 5.0%.

The Talenenergy Martins Crk Llc Williamsport 500-345 kV TX is overloaded to 103.7% of its 1,383 MVA emergency rating for the loss of the Hilltop 500/345kV XFMR. The project contributes 6.7% DFAX (19.1 MW) to this overload.

The Little Chute – Talenenergy Martins Crk Llc Harrisburg 345 kV is overloaded to 103.4% of its 1,626 MVA emergency rating for the loss of the Coronet – Fot Humbug 500 kV. The project contributes 7.1% DFAX (20.4 MW) to this overload.

The Sugar Hill – Ritter Park 345 kV is overloaded to 103.1% of its 1,250 MVA emergency rating for the loss of the Hilltop – Piney Creek Project 500 kV. The project contributes 10.1% DFAX (28.9 MW) to this overload.

The Ft Humbug – Orchard Road Substation 500 kV is overloaded to 102.9% of its 3,014 MVA emergency rating for the loss of the Ft Humbug – W Campus 345 kV. The project contributes 12.7% DFAX (36.3 MW) to this overload.

For the loss of Orchard Rd – Frt Humbug 500 kV line, the Talenenergy Martins Creek Llc Williamsport – Montour Sta 345 kV loads to 102.4% of its 1,130 MVA rating. Project DFAX: 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tellico – Eldorado 138 kV	Rebuild line with higher-capacity conductor	\$10,736,000	100% (\$10,736,000)
Greenwater – Ss3144 Tap 138 kV	Rebuild line with higher-capacity conductor	\$16,949,000	100% (\$16,949,000)
Twisp – Vly Springs 138 kV	Rebuild line with higher-capacity conductor	\$6,871,000	100% (\$6,871,000)
Alder Creek-Bradshaw Substation – Tracy Defense Depot Substation 138 kV	Rebuild line with higher-capacity conductor	\$6,589,000	100% (\$6,589,000)
La Fresa – Munk 138 kV	Rebuild line with higher-capacity conductor	\$2,371,000	100% (\$2,371,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Alder Creek-Bradshaw – La Fresa 138 kV	Rebuild line with higher-capacity conductor	\$8,950,000	100% (\$8,950,000)
Wissota – Berry Street 138 kV	Rebuild line with higher-capacity conductor	\$977,000	52% (\$510,000)
Larson Sreet – Norwich 230 kV	Rebuild line with higher-capacity conductor	\$16,044,000	36% (\$5,833,000)
Firstenergy Albright Substation – Munk 138 kV	Rebuild line with higher-capacity conductor	\$10,005,000	100% (\$10,005,000)
Minco Substation – Larson Street 230 kV	Rebuild line with higher-capacity conductor	\$9,968,000	36% (\$3,576,000)
Wnthrop – Minco 230 kV Line	Rebuild line with higher-capacity conductor	\$14,296,000	46% (\$6,506,000)
Goldsboro – Ironman 230 kV	Rebuild line with higher-capacity conductor	\$65,343,000	36% (\$23,763,000)
Piney Crk Project Substation 500-345 kV Auto	Replace transformer with higher-rated unit	\$31,447,000	39% (\$12,401,000)
Gavin Power Llc – Olympus 230 kV	Reconductor line	\$3,163,000	39% (\$1,221,000)
Fort Humbug – Coronet 500 kV L	Reconductor line	\$49,141,000	40% (\$19,602,000)
Allegheny Energy Units 3 4 & 5 345/1kV TX	Replace transformer with higher-rated unit	\$21,089,000	39% (\$8,275,000)
Allegheny Energy Units 3 4 & 5 Substation 500/1kV Auto	Replace transformer with higher-rated unit	\$33,452,000	39% (\$13,126,000)
Hilltop 500-345 kV Autotransformer	Replace transformer with higher-rated unit	\$23,863,000	40% (\$9,494,000)
Blackrock – Shawville Station 345 kV	Reconductor line	\$16,027,000	40% (\$6,354,000)
Shawville Sta 500/345 kV Transformer	Replace transformer with higher-rated unit	\$26,717,000	100% (\$26,717,000)
Talenenergy Martins Creek Llc Williamsport 500/345kV Transformer	Replace transformer with higher-rated unit	\$21,453,000	40% (\$8,562,000)
Little Chute Substation – Talenenergy Martins Creek Llc Harrisburg 345 kV	Reconductor line	\$6,366,000	40% (\$2,534,000)
Sugar Hill – Ritter Park 345 kV	Reconductor line	\$5,636,000	39% (\$2,212,000)
Ft Humbug – Orchard Road Substation 500 kV	Reconductor line	\$33,972,000	40% (\$13,520,000)
Montour Sta – Talenenergy Martins Creek Llc Williamsport 345 kV	Reconductor line	\$21,755,000	60% (\$13,149,000)
	Total allocated to project		\$239,826,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.